



AMCS2104

Fundamentals of Artificial Intelligence

My Tutorial Book (武林秘籍)

2026/2027 (Session 202605)
Year 2 Semester 1

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Class Detail

Programme	DCS2S1	
Tutorial Group	G5	
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Guideline

1. The class representative or assistant is required to assign one question to each student accordingly (you may follow the name list above) and insert the student's name in the yellow bracket at the end of the assigned question so that the student is aware of their assigned question.
2. Please attempt all the questions before attending the tutorial class as questions without answer may be skipped during discussion. (p/s: no need to be afraid of giving incorrect answers as we will discuss together and learn from each other 🤝)
3. Feel free to include any relevant graphics, images, or video for better understanding.
4. You are always welcome to actively participate in class activities, discussions, and exercises by sharing thoughts and asking questions. (p/s: I won't eat you 😊)
5. Please make an effort to arrive promptly for the tutorial class and strive to attend the required number of sessions.

Quote of the Day

"The beautiful thing about learning is that nobody can take it away from you."

- B.B. King

Table of Contents

Tutorial 1	5
Question 1	5
Question 2	7
Tutorial 2	10
Question 1	10
Question 2	11
Tutorial 3	14
Question 1	14
Question 2	14
Tutorial 4	17
Question 1	17
Question 2	19
Question 3	20
Tutorial 5	22
Question 1	22
Tutorial 6	25
Question 1	25
Tutorial 7	28
Question 1	28
Question 2	29
Question 3	32
Tutorial 8	35
Question 1	35
Tutorial 9	38
Question 1	38
Question 2	41
Tutorial 10	43
Question 1	43
Question 2	44
Question 3	46
Tutorial 11	47
Question 1	47
Question 2	49
Tutorial 12	51
Question 1	51
Question 2	54
Tutorial 13	57
Question 1	57
Question 2	59
Question 3	61

Tutorial 1

Question 1

- a) **Artificial Intelligence (AI)** is a term frequently used to refer to systems equipped with the intellectual capability of humans, such as the ability to reason, discover meaning, generalize, or learn from past experience. However, many experts believe that those systems are still far from as intelligent as humans. Yet, some of those systems outperform humans especially in some cases that involved repetitive or detail-oriented tasks.
- i. A test was proposed in 1950 to determine whether **a machine is truly intelligent**. Name the test and explain how it works. (ANG ZI CHEN)

turing test, The test involves three parties: a computer, a human respondent, and a human interrogator. Direct physical interaction between them is avoided. The human interrogator poses written questions. The computer successfully passes the test if the interrogator cannot tell whether the written responses are coming from a person or the computer

- ii. Is the above-mentioned test still relevant today? Explain the reason why. (BEH CHUN YI)

Turing test is largely considered outdated and not fully relevant today. Because the test is narrow and focuses on deception. The test measures a machine's ability to imitate human conversation, not true understanding, reasoning, or consciousness. Modern AI can pass parts of the test by mimicking human language, but they do not truly think or understand in the human sense.

- iii. **Terminator 2: Judgment Day** is an old but popular science fiction movie. In this movie, **T-800 Terminator** is an **AI character** capable of the intellectual capability of humans portrayed by Arnold Schwarzenegger. Identify whether the **T-800 Terminator** possesses

the following **SIX (6) capabilities**. Justify your answer with the use of an example for each capability. (CHONG WEN KAI)

- Natural language processing
- Knowledge representation
- Automated reasoning
- Machine learning
- Computer vision
- Robotics

- Natural language processing: T-800 can understand human language, English that John Connor speaks and communicates with him.

- Knowledge representation: T-800 knows that T-1000 is hunting for John and remembers it for the whole movie.

- Automated reasoning: Reasons with John to assure him that he is a terminator because he doesn't bleed.

- Machine learning: T-800 learns that T-1000 can't die easily to attacks and bullets but can be stalled, so T-800 keeps shooting or attacking him to disrupt his attacks or movements.

- Computer vision: From what he sees, T-1000 is at the edge of the platform with molten steel below, so he shot him with the grenade launcher. The T-1000 loses balance and falls from the platform and into molten steel.

- Robotics: Can drive motorcycle without license and go vroom vroom.

iv. Predict if **T-800 Terminator** can **pass the test** mentioned in **Question 1 a) i)**. Justify your answer. (NEO JIA LER)

Yes, because T-800 Terminator can communicate naturally, learn from experience and imitate human behavior effectively, making it difficult to identify it as a machine.

b) Overall, there are many different approaches to AI and they can be divided into **FOUR (4)** categories.

i. Name the **FOUR (4)** categories. (CLEMENT HO JUN HAO)

- 1) Thinking humanly
- 2) Acting humanly
- 3) Thinking rationally
- 4) Acting rationally

ii. Describe how the **FOUR (4)** categories are related to each other with the use of an illustration (e.g., Table). (ENG ZI SHAN)

	Humanly	Rationally
Thinking	-AI tries to think like a human -Focuses on how humans learn, reason, remember, and solve problems. -to simulate the human thought process. -using cognitive modelling approach -eg. Intelligent tutoring system that adapts to a student's thinking.	-AI tries to think logically. -Uses formal logic and reasoning rules to solve problems. -using Law of Thought approach -eg. Rule-based Tax Calculation System applies predefined tax rules and formulas to calculate the correct amount of tax based on a person's income and deductions.
Acting	-AI tries to behave like a human. -to make its behaviour indistinguishable from a human -using turing test approach -eg.Voice assistant such as Siri or Google Assistant.	-AI tries to take the best action to achieve a goal. -Consider available information before making decisions. -To maximize success and achieve the best outcome. -using rational agent approach

		-eg. Robot vacuum that finds the most efficient cleaning path.
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iii. One of the categories requires scientists to **understand how humans think**. (LEE NI)

- a) Name the A.I. approach.
- b) Then, name **THREE (3)** ways where it can be done, and briefly describe each of them.

- a) The Cognitive Modelling Approach
- b)
 - Introspection: Try to catch our own thoughts as they go by
 - Psychological experiments: Observe a person in action
 - Brain imaging: Observe the brain in action

iv. One of the categories requires "**right thinking**" that leads to "**correct inferences**". (FIONNA CHAN PUI YEE)

- a) Name the A.I. approach.
- b) Describe **ONE (1)** challenge of this approach.

- "Laws of thought" approach
- It is extremely difficult to express informal, uncertain, or real-world knowledge strictly in the formal logical terms required by this method.

v. One of the categories requires a computer to **pass a special test** proposed in 1950. (JONG HUI XUAN)

- a) Identify the original name of the test.
- b) Describe the main purpose of the test.

- A) Turing Test
- B) Test a computer's ability to exhibit intelligent behavior equivalent to human.

Question 2

- a) The idea of intelligence, which is mainly concerned with **rational action** is widely adopted by the AI community. In this course, the **Rational Agent approach** is also adopted, in which **rational agents** are referred to as **intelligent agents**.
- i. Explain why the **Rational Agent approach** is widely used. (PANG YI WEN)

The Rational Agent approach is widely adopted because it shifts AI from philosophy to practical engineering. Instead of asking if a machine can "think," it asks: "Does it take the best action to achieve its goal?" First, it provides an **objective measure of success**. Human intelligence is too subjective to test, but a Rational Agent focuses on **maximizing a clear performance measure**, giving engineers a concrete standard to optimize.

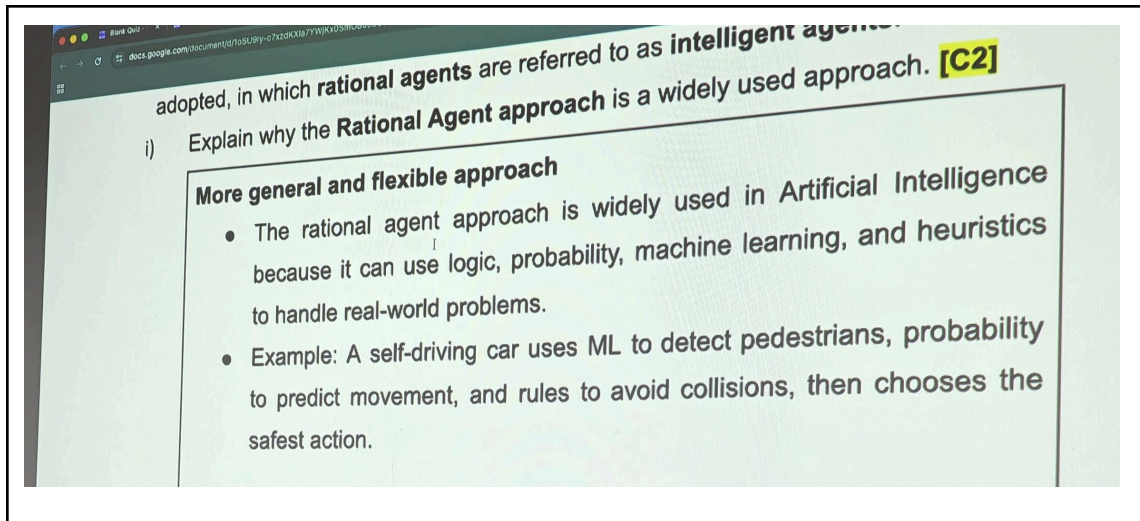
Second, it focuses on actions, not internal thinking. In engineering, the final output is what matters. By focusing strictly on the Percept-Action loop, we do not need to replicate the complex human brain; we only need the machine to do the right thing.

Third, it avoids human limitations. Humans get tired, panic, and make emotional mistakes. A Rational Agent aims for optimal correctness based on logic and data, which is essential for high-stakes fields like aviation or medicine.

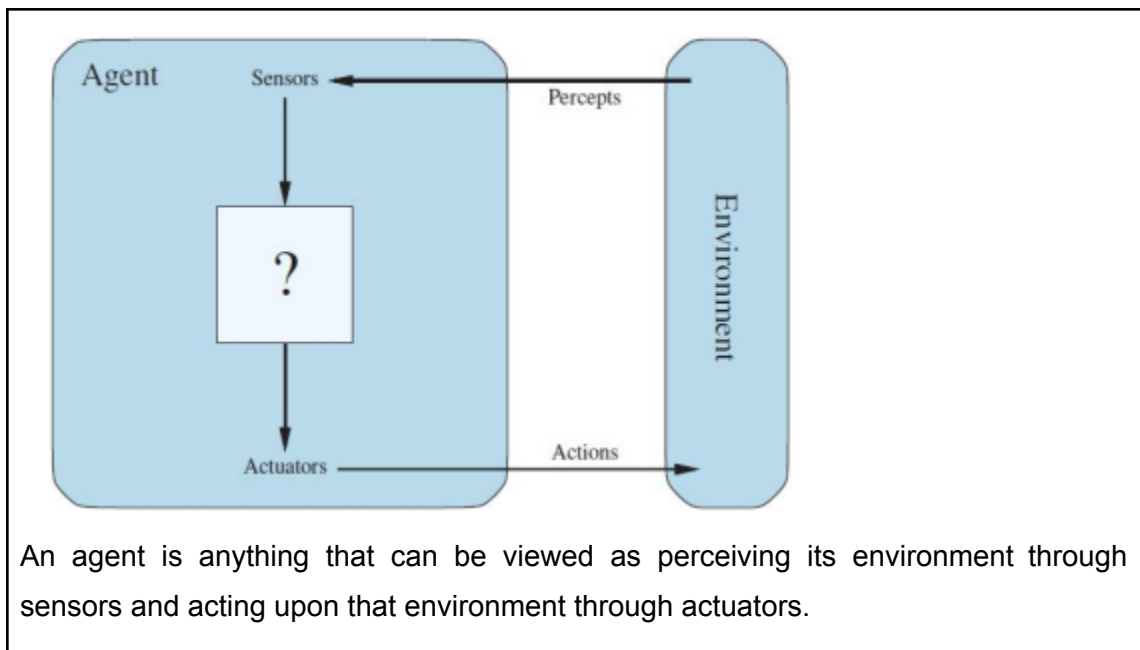
Finally, it is universal. The same basic framework works for a simple thermostat or a complex autonomous drone, providing a unified paradigm for the entire AI community.

Scientifically well-defined and measurable

- It is widely used because intelligence is defined using a clear performance measure, making it easy to evaluate and optimize.
- Example: A robot vacuum cleaner is judged by how much area it cleans, time taken, and battery used, and it chooses actions that maximize cleaning efficiency.



ii. With the aid of a diagram, explain what an **agent** is. (TAN HAO KIT)



iii. Differentiate between an **agent function** and an **agent program**. (KHOO CHEE YOUNG)

Agent Function

- An abstract mathematical description that maps any given percept sequence to an action.

Agent Program

- The actual code running on that architecture. It maps the current percepts to a specific action.

iv. Define **rational agents**. (NG YIT KAI)

Rational agents are defined as agents that act to **achieve the best expected outcome** for a given performance measure, based on their current percepts and built-in knowledge.

v. Based on the definition provided in **Question 2 a) iv)**, identify and briefly describe **THREE (3)** capabilities of **rational agents** that are lacking in **agents** (please refer to the answer provided for **Question 2 a) ii)**). (LEN CHUN HOE)

Expected Utility Maximization

Rational agents select actions that maximize their expected utility based on probabilities and preferences, even under uncertainty.

Goal-Directed Deliberation and Planning

Rational agents maintain internal models of the world and actively deliberate to form plans that achieve specific goals, considering sequences of actions.

Learning and Adaptation from Experience

Rational agents can improve their performance over time by updating their knowledge, models, or utility estimates based on new percepts and outcomes.

b) **Figure 1** below illustrates a vacuum agent that is present in a made-up vacuum world.

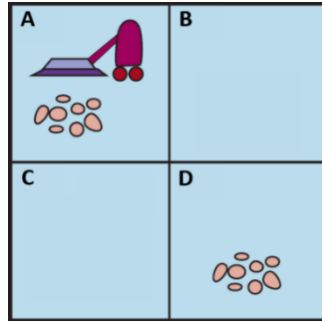


Figure 1. A simple vacuum world

- i. Briefly describe its possible **percept(s)** and **action(s)**. (LEONG ZHENG YANG)

Percepts (sensors detect):

- Location – which room the agent is in (A, B, C, or D).
- Dirt status – whether the current room is clean or dirty.

Actions (actuators can do):

- Move Left / Move Right / Move Up / Move Down (depending on grid layout).
- Suck – to clean the current room if dirty.
- NoOp – do nothing (if all rooms are clean and task finished).

- ii. Suggest an **agent function** for the vacuum agent. (LUI FANG YEE)

If dirt is detected, the agent immediately cleans it.
Otherwise, it continues moving to search for dirt.

- iii. Based on the information provided below, identify whether the agent is a **rational agent**.

Justify your answer. (NATALIE ANG XI QI)

- The performance measure awards one point for each clean square at each time step, over a lifetime of 1000-time steps.
- The geography of the environment is known a priori, but the dirt distribution and the initial location of the agent are not.

- The agent is capable of moving to the left, moving to the right, and sucking (to clean the dirt).
- The agent is equipped with a reliable location and dirt sensor.

Yes, the agent can be a rational agent.

It aims to maximize the performance measure by keeping as many squares clean as possible over 1000 time steps. Although the dirt distribution and starting location are unknown, the agent has reliable location and dirt sensors, allowing it to accurately perceive its environment. It can move left, move right and suck dirt when detected. By always choosing the appropriate action based on its current percepts, the agent maximizes its expected performance. Therefore, it satisfies the definition of a rational agent.

No

Under these circumstances, the agent's expected performance will be worse than any other agent's. The reason is that it is only capable of moving to the left and right, and hence it is only capable of cleaning two squares (out of four squares). In conclusion, the agent is not rational.

- iv. Define a **performance measure**. Then, explain why **performance measures** are important in the design of a rational agent. (RANAN GOH ZHI LIN)

A possible measure from Figure 1 could be "Number of dirt squares cleaned within a fixed number of time steps".

Performance measures are important because they define success or rationality. A rational agent is one that acts to maximize its expected performance according to the performance measure. Without a clear performance measure, it is impossible to say whether an agent is acting rationally.

For example, if a vacuum robot's measure is "number of dirt squares cleaned in 5 minutes," it will prioritize dirty areas rather than wandering aimlessly.

Tutorial 2

Question 1

- a) Environments that agents exhibit can be categorised into 12 types as listed in **Table 1**:
- i. Briefly describe each of them. (RONAN CHAN A/L HARICHAN)

Fully Observable

- The agent's sensors give it complete, up-to-date access to environment's entire state

Partially Observable

- The agent's sensors are limited or noisy, it cannot see the full state

Single-agent

- Only one autonomous agent operates in the environment

Multi-agent

- Multiple autonomous agents operate simultaneously, potentially interacting or competing

Deterministic

- The next state of the environment is entirely determined by the current state and the agent's action

Non-deterministic

- The same action in the same state can lead to different outcomes

Episodic

- The agent's experience is divided into independent episodes, the current action does not affect future situations

Sequential

- The agent's current action affects all future states, long-term planning is required

Static

- The environment does not change while the agent is deliberating

Dynamic

- The environment can change while the agent is thinking

Discrete

- The environment has a finite number of actions or states

Continuous

- The environment can have an infinite number of states

- ii. Provide **ONE (1)** example for each type of environment. (WONG JIA JIAN)

Table 1: 12 types of environment's properties

Fully Observable	Episodic
Partially Observable	Sequential
Single-agent	Static
Multi-agent	Dynamic
Deterministic	Discrete
Non-deterministic	Continuous

Fully Observable - Chess game	Episodic - Mail sorting system
Partially Observable - Card game	Sequential - Chess game
Single-agent - Crossword puzzle	Static - Crossword puzzle
Multi-agent - Taxi driving	Dynamic - Soccer
Deterministic - Tic-Tac-Toe game	Discrete - Tic-Tac-Toe game
Non-deterministic - Soccer	Continuous - Basketball

- b) **Meta**, formerly known as Facebook, is one of the social media companies that makes use of agent, referred to as the **facial recognition system**, to automatically identify people in images.

- i. Construct a **PEAS** description for the problem that the agent is going to solve. (Yap Jia Yu)

Performance measure - recognize the right person do it fast and don't mistake

Environment - the billions of photos and videos uploaded to facebook and instagram.

Billion of global users with different faces, angles, and lighting.

Actuators - sending notifications alerting you someone posted a photo that might be you.

Sensors - the uploaded images or videos themselves. The AI uses this to narrow down its guesses

Performance measure	Accuracy of recognizing human face
Environment	Images / photographs
Actuators	Display screen (to display the scanned image recognized faces)
Sensors	Keyboard, mouse

- ii. Then, identify the **properties** of the environment(s). Justify your answer. (ANG ZI CHEN)

Fully Observable — The agent's sensor gives it access to the complete state it needs at the moment of classification. The whole image is available at once, so the agent does not need to maintain an internal state to track hidden parts of the world.

Single-Agent — Only one agent operates on the task. There are no other agents whose behaviour the system must compete or cooperate with to maximise its performance measure; identifying a face in an image does not depend on another agent's actions.

Deterministic — The next state of the environment is completely determined by the current state and the agent's action. Given the same image, the system's identification produces the same result, and the act of classifying does not introduce randomness into the environment.

Episodic — The agent's experience is divided into atomic episodes: each image is processed independently. The system receives one percept, performs a single action, and the outcome does not depend on previous images. The agent therefore does not need to think ahead across episodes.

static— The image itself does not change while the agent is deliberating on it, so the environment is static in that sense.

Continuous — Images are made up of pixel intensity values spanning a continuous range, giving effectively infinite possible states and percepts. The percept space is high-dimensional and continuous rather than a finite set of discrete states.

- c) **Hearts**, which used to be part of the older version of the Windows operating system, is a famous trick-taking **card game** (<https://www.free-hearts.com/#/app/game>).
- i. Construct a **PEAS** description for the problem that the agent is going to solve. (BEH CHUN YI)

Performance measure- Minimize penalty points, win with lowest score, successfully shoot the moon when advantageous, play only legal moves.

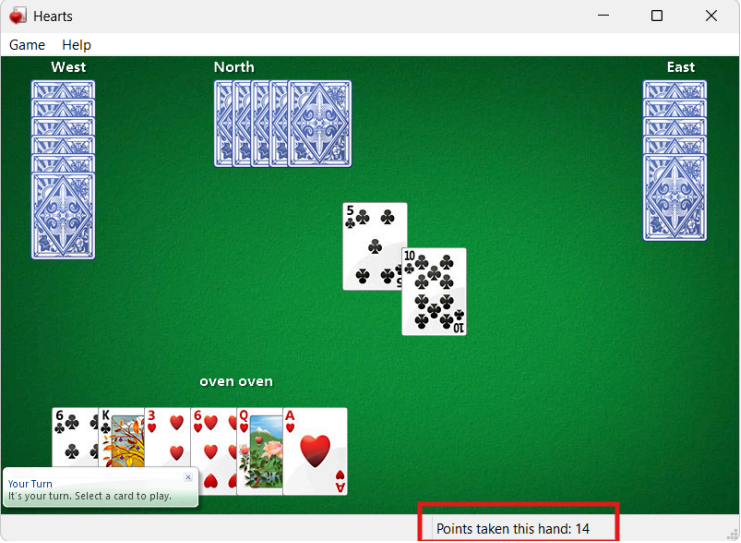
Environment- Standard 52-card deck, 3 opposing players, current trick and round, passing phase, score limit (e.g., 100 points)

Actuators-Play a card, select and pass 3 cards, submit move via game interface

Sensor-Its own hand of cards, the cards played in the current round, past cards that were played, cards it got when passing, each player's score, whose turn it is, and whether Hearts have been played yet.

- ii. Then, identify the **properties** of the environment(s). Justify your answer. (CHONG WEN KAI)

- Partially observable. Because the agent cannot see other's cards, the agent has to observe its own card and also keep track of cards played.
- Multiagent. The agent is playing against 3 other agents so this is a competitive play of multiple agents.
- Stochastic. This is non-deterministic because agents are uncertain about cards that would get played, due to each player getting 13 random cards at start. Example, playing 3 of clubs didn't earn points in the last match, but earned points in the current match.
- Sequential. When the agent plays a card, it affects how many points they might get or could get in later turns in total. Which determines who wins.



- Static. The cards won't suddenly change so agents can perceive the environment to act.
- Discrete. The possible actions of cards are finite, so are the number of cards played or points taken.
- Known. Rules and strategies of Hearts are known. Example: The player with the lowest points wins.

- a) An intelligent **Chinese Chess** programme that can give you an enjoyable and satisfactory gameplay:
- Construct a **PEAS** description for the problem that the agent is going to solve. (CHOW ZHEN WAI)

Performance Measure

The agent aims to win the game by checkmating the opponent while making strong strategic decisions. It should also provide an enjoyable and challenging gameplay experience by selecting moves efficiently and at an appropriate difficulty level.

Environment

The environment is the Chinese Chess board, including all pieces, legal moves, game rules, and the opponent (human or another AI). The board state changes after each move made by either player.

Actuators

The agent performs actions by selecting and executing legal chess moves, such as moving a piece from one position to another. It may also offer features like resigning, requesting a draw, or adjusting the difficulty level.

Sensors

The agent observes the current board configuration (via camera for physical board, or board state notation for digital board), the positions of all pieces, the opponent's moves, and the game status (e.g., check, checkmate, or draw). This information is used to determine the next move.

- ii. Suppose the environment is the chess game; identify the **properties** of the environment(s). Justify your answer. (NEO JIA LER)

Fully Observable-The AI can observe every move made by the opponent and knows the exact positions of all pieces on the board.

Multiagent-The AI must consider the strategies and actions of the opponent while making decisions.

Deterministics- Chess game with no element of chance or randomness.

Sequential-The AI must consider the consequences of current actions on future game states.

Static-The chessboard remains unchanged until either the AI or the opponent makes a move

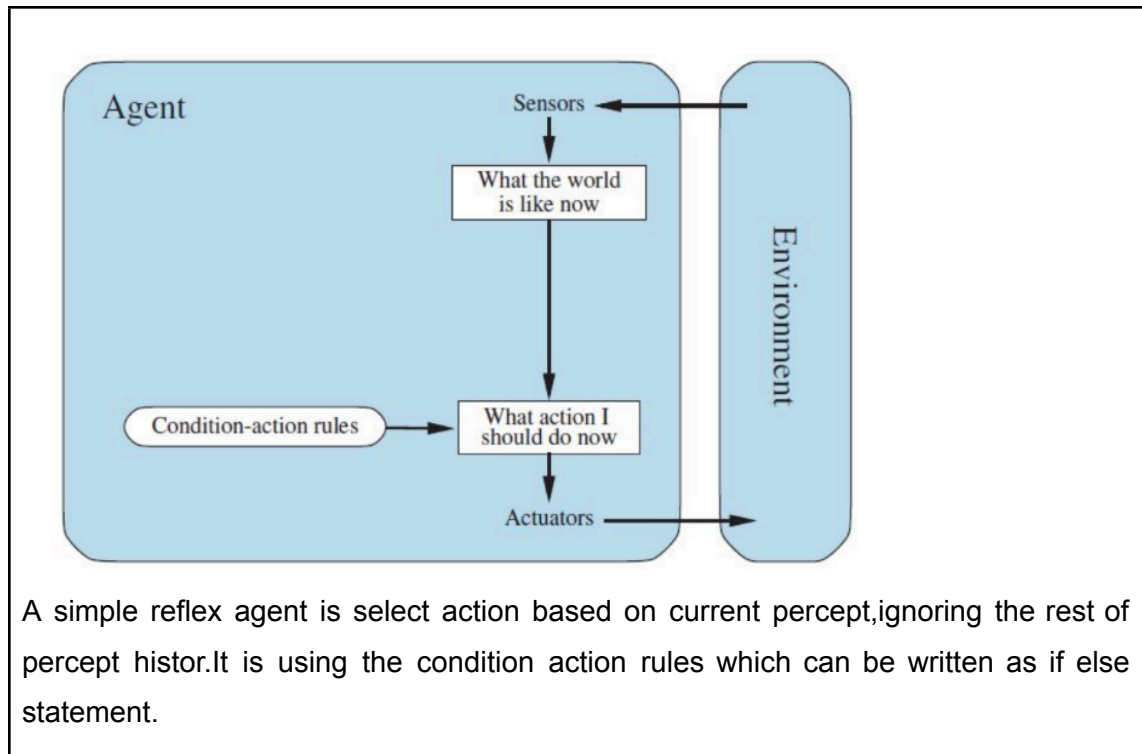
Discrete-The board consists of fixed squares and moves occur in distinct turns rather than continuously.

Known-The AI has complete knowledge of the game rules and can predict the effects of every legal move.

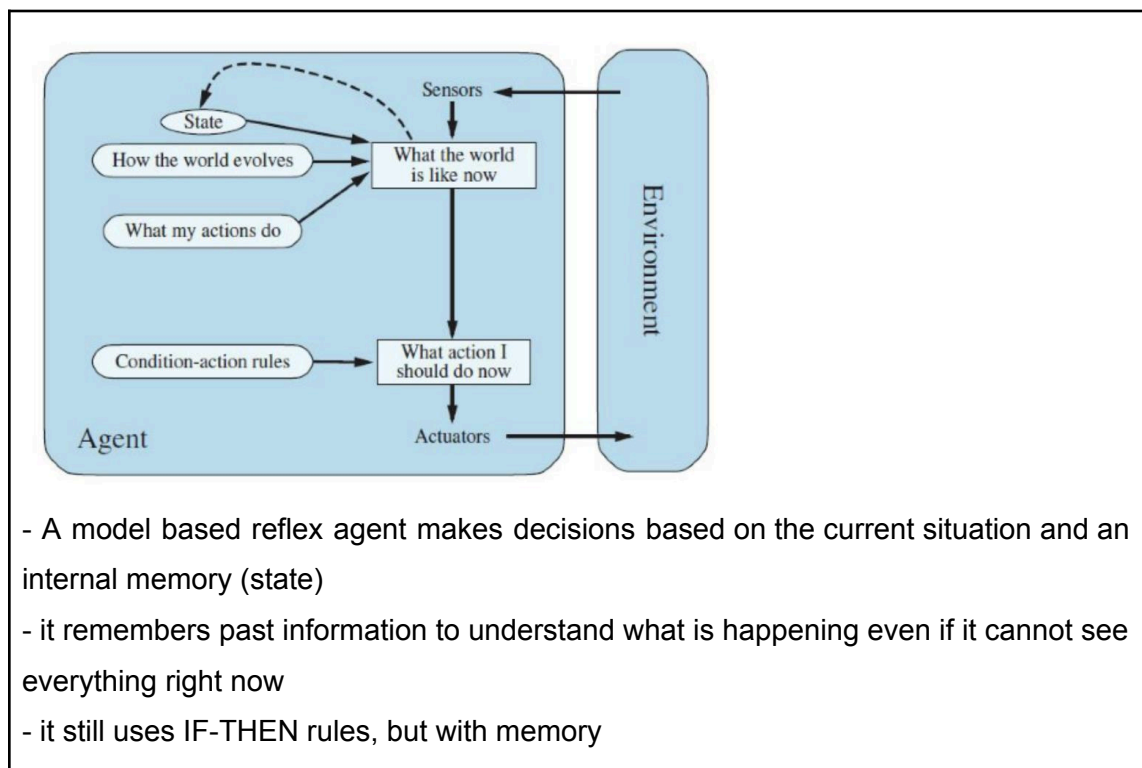
Question 2

- a) With the use of the schematic diagrams provided in the lecture notes or main reference, explain how an agent that is implemented based on the following agent programs make decision on their actions:

i. Simple reflex agents (CLEMENT HO JUN HAO)

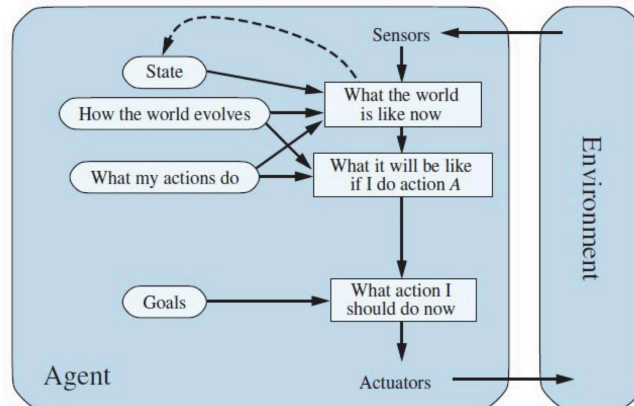


ii. Model-based reflex agents (ENG ZI SHAN)



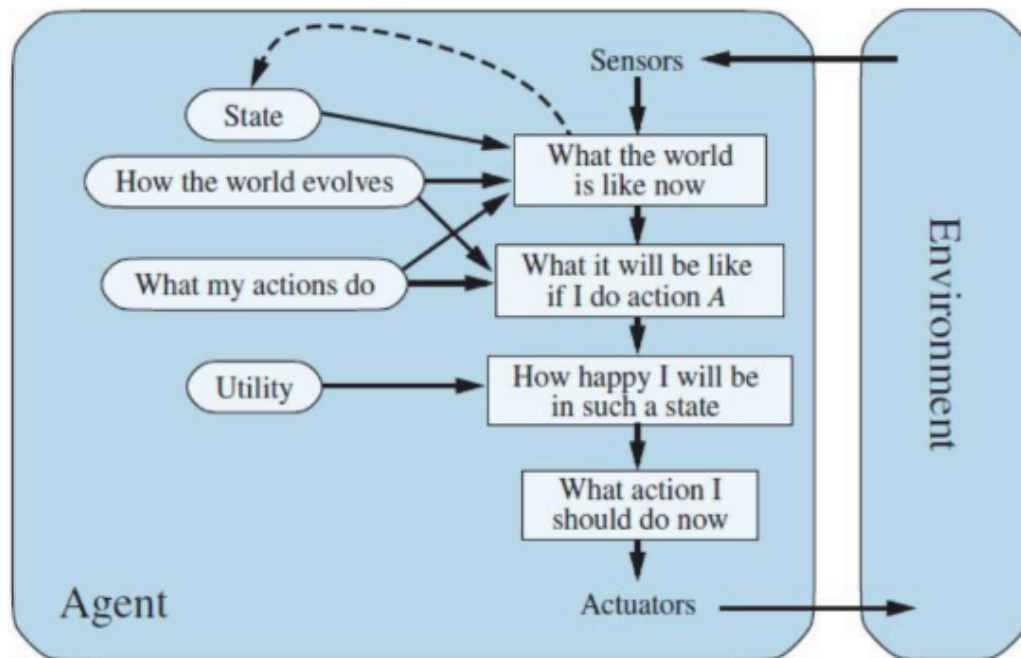
- it reacts+remembers, but does NOT plan a full strategy.

iii. Goal-based agents (LEE NI)



A goal-based agent makes decisions based on a specific goal it wants to achieve. It does not only react to the current situation, but also plans and predicts the outcome of different actions. It chooses the action that brings it closer to achieve its goal.

iv. Utility-based agents (FIONNA CHAN PUI YEE)



- Chooses the action that leads to the best expected utility by averaging over all possible outcome states weighted by the probability of the outcome.
- utility function assigns score to different world states according to exactly "how happy they would make the agent".
- If the internal utility function and the external performance measure are in agreement, then a rational agent will choose actions that maximize its utility.

b) Suppose two agents in each of the following pairs are designed to solve the same problem. For each pair of them, identify one agent program that is more suitable to be applied as the solution, and justify your answer:

- Table-driven agents vs. Simple reflex agents (JONG HUI XUAN)

Simple reflex agents are more suitable to be applied as the solution because of its memory efficiency because they do not store history. A Simple Reflex Agent makes decisions based on the current percept and a set of conditions–action rules (if–then rules). It responds immediately to the current situation without searching through a large table..

- Simple reflex agents vs. Model-based reflex agents (PANG YI WEN)

Model based reflex agent is more suitable to be applied as the solution because model based reflex agent will combined the previous experience and it have a internal tracking. While the simple reflex agents is just suitable to solve the domain problem because it doesn't have history .

- Model-based reflex agents vs. Goal-based agents (TAN HAO KIT)

Model-based reflex agent relies strictly on predefined condition actions rules applied to its current and past environment, whereas a goal-based agent uses planning, search algorithm and foresight to choose paths that actively achieve a specific objective.

- Goal-based agents vs. Utility-based agents (KHOO CHEE YOUNG)

- Goal-based agents focus on achieving a specific, defined objective regardless of the effort required.
Example: GPS finding route to the destination.
- Utility-based agents use a numerical score, or "utility," to evaluate different actions and select the one that maximizes the desirability of the outcome.
Example: GPS finding the fastest route to the destination .

Tutorial 3

Question 1

- a) A **learning agent** in artificial intelligence is a type of agent equipped with learning capabilities and can learn from its past experiences.
- i. Differentiate a learning agent from a non-learning agent in **THREE (3)** ways. (NG YIT KAI)

1. Learning from experience

- Learning agent: A learning agent can learn from its past experiences and improve its performance over time.
- Non-learning agent: A non-learning agent cannot learn from experience and only follows predefined rules.

2. Adaptability

- Learning agent: A learning agent can adapt its behaviour when the environment changes.
- Non-learning agent: A non-learning agent cannot adapt to changes and continues using the same fixed rules.

3. Performance improvement

- Learning agent: A learning agent continuously improves its accuracy and efficiency through learning.
- Non-learning agent: A non-learning agent's performance remains the same because it cannot learn or improve.

- ii. Explain the importance of empowering an agent with **learning capabilities**. (LEN CHUN HOE)

It can learn from past experiences and continuously improve its performance. A learning agent can adapt to changes in the environment, make better decisions based on its learned knowledge, and reduce the need for manual programming. Therefore, the agent can solve problems more efficiently, accurately, and effectively.

- b) In general, a learning agent can be divided into **FOUR (4)** conceptual components.
- i. Name the **FOUR (4)** conceptual components. Then, briefly describe each of them. (LEONG ZHENG YANG)

Learning Element – Responsible for making improvements to the agent's knowledge/behavior over time. It uses feedback from the critic to modify the performance element so the agent does better in the future.

Performance Element – Responsible for selecting external actions. This is essentially the “agent” itself as previously conceived — it takes in percepts and decides on actions, using whatever knowledge/policy it currently has.

Critic – Tells the learning element how well the agent is doing with respect to a fixed performance standard. It evaluates the outcomes of actions (based on external feedback or defined performance measures) and provides feedback that the learning element uses to make adjustments.

Problem Generator – Responsible for suggesting actions that will lead to new and informative experiences, even if they aren't necessarily optimal in the short term. It encourages exploration so the agent doesn't just keep repeating known-safe actions and can discover better strategies.

- ii. Provide an example scenario that can be used to explain how each of these components works collaboratively in making an agent learn. (LUI FANG YEE)

Self-Driving Car

Performance Element

- The self-driving car observes the road and decides whether to accelerate, brake, or turn.

Critic

- The critic evaluates the car's performance, such as whether it stayed in its lane, avoided collisions, and reached the destination safely.

Learning Element

- Based on the critic's feedback, the learning element updates the driving strategy so the car makes better decisions in future trips.

Problem Generator

- The problem generator encourages the car to explore new driving routes or strategies (when safe), allowing it to gain new experience and improve its driving ability.

Question 2

- a) The process of solving a search problem by a problem-solving agent can be divided into **FOUR (4)** phases, of which one of them is **problem formulation**.

- i. Identify the other **THREE (3)** phases. (NATALIE ANG XI QI)

Goal Formulation, Solution Formulation and Solution Execution.

- ii. Briefly explain the purpose of each of the **FOUR (4)** phases. (RANAN GOH ZHI LIN)

1. Goal Formulation - Decide what goal the agent wants to achieve.

2. Problem Formulation - Define the problem by identifying the initial state, actions, goal state, and path cost.

3. Search - Find the best path or sequence of actions to reach the goal.
4. Execution - Perform the chosen actions to achieve the goal.

b) A search problem can be defined in terms of **SIX (6)** elements.

i. Identify the **SIX (6)** elements. (RONAN CHAN A/L HARICHAN)

- 1) States
- 2) Initial State
- 3) Goal State
- 4) Action
- 5) Transition Model
- 6) Cost Function

ii. Briefly describe the **SIX (6)** elements. (WONG JIA JIAN)

States: A set of possible states that the environment can be in, referred to as state space.

Initial state: A state that the agent starts in.

Goal state: A state that represents or satisfies the goal.

Action: A set of possible actions available to the agent in a particular state.

Transition model: A description of what each action does.

Cost function: The cost of applying an action in state to reach another state.

iii. Formulate the following **sliding puzzle problem** in terms of the **SIX (6)** elements. (Yap Jia Yu)



Figure 1. A smiley sliding puzzle

State: a state description specific the location of each of the tile

Initial state: any state can be designated as the initial state

Goal state: any state could be the goal.

- One of the most commonly used goal state is specified in the image provided above

Action: • The movement of the blank space: Left, Right, Up, or Down.

- If the blank is at an edge or corner then not all actions will be applicable

Transition model: • It maps a state and action to a resulting state.

- For example, if we apply Down to the state specified in the image provided above. the resulting state has the blank moves down

Cost function: $A1 > A2$ cost 1

iv. Explain the relationship between the following pairs of terms: (ANG ZI CHEN)

- Action vs. Action-cost function
- Action vs. Path
- Path vs. Path cost
- Path cost vs. Action-cost function
- Path vs. Solution
- Optimal solution vs. Solutions

Action is the agent can move from one state to another
Action-cost function is how much it cost of doing that action

Action a single step.
Path is a sequence of action joined together

Path is a sequence of actions from start to end
Path cost is a total cost of the path

Action-cost function is the cost of a single action

Path cost is the cost is calculated by adding up the action costs

Paths start at the initial state and goal state.

Every solution is a path, but some paths don't reach the goal

A problem may have many solutions

An optimal solution is a solution with the lowest path cost.

c) **Abstraction** refers to the process of removing detail from a representation.

i. Identify **TWO (2)** properties of abstraction. Then, briefly describe each of them. (BEH CHUN YI)

Validity - Any abstract solution must be expandable into a real, valid solution in the original, detailed problem.

Usefulness- Solving the abstract problem must be significantly easier than solving the original detailed problem directly

ii. Identify **TWO (2)** examples of abstraction for the sliding puzzle problem formulated in Question 1) b) iii). (CHONG WEN KAI)

Following sliding puzzle problem in text



Figure 1. A smiley sliding puzzle

Initial state abstraction: A2

Action abstraction: $A2 > A1$ ✗

- Actions available are abstracted into the movement of the blank tile instead of the

tiles.

- Actions are abstracted to their beginning and final states, ignoring the intermediate locations where the tile is sliding.

Tutorial 4

Question 1

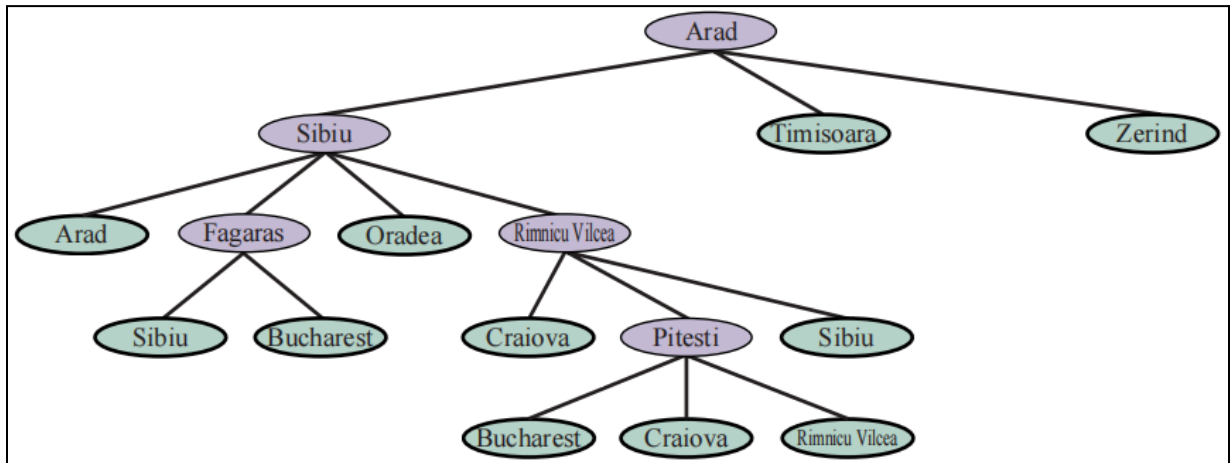


Figure 1. A search tree

a) **Figure 1** shows a search tree generated by A* search while searching a path from Arad to Bucharest.

i. Define **loopy paths** and **redundant paths**. (CHOW ZHEN WAI)

loopy path is a path that contain a repeated state, meaning it revisits a state/node that has already been visited earlier in the same path, forming a cycle.

redundant path is a path that reaches a state that has already been reached by another path, resulting in unnecessary repeated exploration

ii. Highlight any loopy paths and redundant paths that are present in the search tree in **Figure 1**. Then, identify each of them as either a loopy path or a redundant path. (NEO JIA LER)

Loopy Paths

-Arad → Sibiu → Arad

-Arad → Sibiu → Fagaras → Sibiu
 -Arad → Sibiu → Rimnicu Vilcea → Sibiu

Redundant Paths

-Arad → Sibiu → Fagaras → Bucharest
 -Arad → Sibiu → Rimnicu Vilcea → Pitesti → Bucharest
 -Arad → Sibiu → Rimnicu Vilcea → Craiova
 -Arad → Sibiu → Rimnicu Vilcea → Pitesti → Craiova

- iii. Suggest **ONE (1)** way to solve the issue caused by the loopy path and redundant path.
(CLEMENT HO JUN HAO)

Transition from a tree search to a graph search .Before expanding a node from the frontier (open list), check if its state is already in the explored set. If it is, simply discard the node rather than expanding it again.

- b) **Node expansion** and **node generation** are two important processes that happen while a search tree is being constructed. Discuss **TWO (2)** differences between them. **(ENG ZI SHAN)**

Node expansion	Node generation
Node expansion is the process of applying all available actions to the current node to determine all possible next states.	Node generation is the process of creating new child nodes from those states and adding them to the frontier so they can be explored later.
The purpose of node expansion is to discover all possible successor states from the current node.	The purpose of node generation is to create and store those successor nodes in the frontier, allowing the search algorithm to continue exploring them later.

- c) Both **frontier** and **reached** are **TWO (2)** important concepts used to implement search algorithms.
- i. Differentiate the **frontier** from the **reached**. **(LEE NI)**

Frontier

- Frontier (or open list) refers to a set of unexpanded nodes that reside in the memory. The root node is first added to the frontier, and then it is removed from the frontier for expansion.

Reached

- Any state that has had a node generated (whether or not that node has been expanded) for it has been reached. Nodes that have already been expanded are referred to as the explored set (or closed list).

ii. Discover all elements stored in the **frontier** and **reached** per the search tree in **Figure**

1. (FIONNA CHAN PUI YEE)

Frontier

- **Arad** (the leaf under Sibiu)
- **Sibiu** (the leaf under Fagaras)
- **Bucharest** (the leaf under Fagaras)
- **Oradea** (the leaf under Sibiu)
- **Craiova** (the leaf under Rimnicu Vilcea)
- **Bucharest** (the leaf under Pitesti)
- **Craiova** (the leaf under Pitesti)
- **Rimnicu Vilcea** (the leaf under Pitesti)
- **Sibiu** (the leaf under Rimnicu Vilcea)
- **Timisoara** (the leaf under the root Arad)
- **Zerind** (the leaf under the root Arad)

Reached

Arad (Root)

- **Sibiu**
- **Fagaras**
- **Rimnicu Vilcea**
- **Pitesti**

- iii. A frontier that is implemented with the use of an appropriate data structure (e.g., a **queue**) can help improve the efficiency of a search algorithm. Briefly describe **THREE (3)** types of queue that are usually used to implement a frontier are listed below: (JONG HUI XUAN)

- **First in first out queue**
- **Last in first out queue**
- **Priority queue**

- **First-In, First-Out (FIFO) Queue:** Elements are removed strictly in the order they were added. The first node placed into the frontier is the first one to be extracted and expanded. This is primarily used in Breadth-First Search (BFS) to explore the tree layer by layer.
- **Last-In, First-Out (LIFO) Queue (Stack):** The element added most recently is the first one to be removed. This structure causes the search to explore deeply down a single branch before backtracking, which is the foundational mechanism for Depth-First Search (DFS).
- **Priority Queue:** Each element is stored along with an associated numerical priority or cost value. Elements are dequeued based on this priority (typically the node with the lowest cost is popped first). This is used in informed searches like A* Search and Uniform Cost Search (UCS).

- iv. Suppose A* search expands nodes (see **Figure 1**) in the sequence of **Arad > Sibiu > Rimnicu Vilcea > Fagaras > Pitestia**, identify an appropriate type of queue used to implement the frontier of A* search. Justify your answer. (PANG YI WEN)

Priority Queue

Evaluation Function Valuations ,A* search guides it's exploration using the evaluation function $f(n) = g(n) + h(n)$. $g(n)$ is the cost from the start node to node n , and $h(n)$ is

the estimate cost from n to the goal

- v. Suppose the frontier of the search tree in **Figure 2** is implemented as a **first in first out queue** and **last in first out queue**. Then, assuming that nodes are expanded from left to right, for each queue, identify the sequence of nodes in the frontier in a correct order as the tree is constructed from the beginning until the end. (TAN HAO KIT)

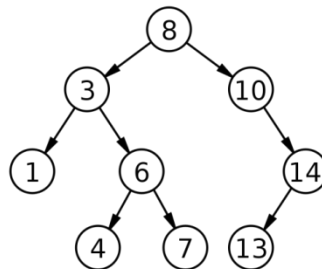


Figure 2. A search tree

[T4Q1\(c\)\(iv\)_InteractiveBoard](#)

FIFO

[8]

[3, 10]

[10, 1, 6]

[1, 6, 14]

[6, 14]

[14, 4, 7]

[4, 7, 13]

[7, 13]

[13]

LIFO

[8]

[3, 10]

[1, 6, 10]

[6, 10]

[4, 7, 10]

[7, 10]

[10]

[14]

[13]

Question 2

a) The performance of search algorithms can be compared in **FOUR (4)** ways in terms of **Completeness**, **Optimality**, **Space complexity**, and **Time complexity (COST)**.

i. Provide a brief description for each element of **COST**. (KHOO CHEE YOUNG)

Completeness measures whether a search algorithm is guaranteed to find a solution if at least one valid solution exists.

Optimality determines whether the algorithm finds the best possible solution among all existing solutions.

Space complexity evaluates the amount of memory the algorithm requires in the worst case scenario to execute the search

Time complexity measures how many execution steps the algorithm takes to find a solution.

ii. **Big-O notation** is used to measure time and space complexities. Briefly describe big-O notation. Then, list down different types of complexities. (NG YIT KAI)

Big-O notation describes the upper bound of an algorithm's growth rate as input size(n) increases. It expresses how the time or space requirements scale, ignoring constants and lower-order terms, so it focuses on worst-case performance as input size(n) grows large.

Different types of complexities

- $O(1)$
- $O(\log n)$
- $O(n)$

- $O(n \log n)$
- $O(n^2)$
- $O(n^3)$
- $O(2n)$
- $O(n!)$

- iii. Identify all possible outcomes of the measurement for each element of **COST**. Then, indicate at most **TWO (2)** preferred outcomes among all possible outcomes for each element. (LEN CHUN HOE)

Completeness

possible outcomes: Complete / Not complete

preferred: Complete

Optimality⁴

possible outcomes: Optimal / Not optimal

preferred: Optimal

Space complexity

possible outcomes: $O(1)$, $O(n)$, $O(\log n)$, $O(n \log n)$, $O(n^2)$, $O(n^3)$, $O(2n)$

preferred: $O(1)$, $O(\log n)$

Time complexity

possible outcomes: $O(1)$, $O(n)$, $O(\log n)$, $O(n \log n)$, $O(n^2)$, $O(n^3)$, $O(2n)$

preferred: $O(1)$, $O(\log n)$

Question 3

- a) With the use of a map (see **Figure 4**) that contains 10 cities (where the distance between any two cities is indicated by a number in brown) as a state space, find **the shortest path** to travel from city **A** to city **J** using **breadth-first search**.

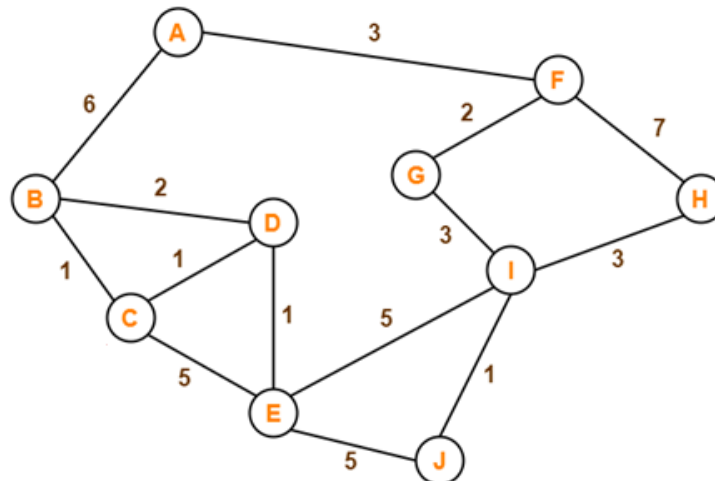


Figure 4. A map

Suppose **breadth-first search** is implemented with a **graph search algorithm**, the **early goal test** is applied, and nodes are expanded in alphabetical order:

- i. Briefly describe what an **uninformed search** is. (LEONG ZHENG YANG)

An uninformed search is a search strategy that explores the state space without any additional information about how close a state is to the goal. It only knows the problem definition — the start state, goal state, successor function, and path cost — but has no heuristic to guide it toward the goal more efficiently. Examples include BFS, DFS, and Uniform Cost Search.

- ii. Formulate the above path-finding problem. (LUI FANG YEE)

1. Initial State

City A (the starting city)

2. Actions

Move from the current city to any adjacent (connected) city.

Example:

From A $\rightarrow$ {B, F}

From B $\rightarrow$ {A, C, D}

From E $\rightarrow$ {C, D, I, J}

(and so on for all connected cities)

3. Transition Model (RESULT(s, a))

Applying an action moves the agent from one city to the connected city.

Example:

$\text{RESULT}(\text{In}(\text{A}), \text{Go}(\text{F})) = \text{In}(\text{F})$

$\text{RESULT}(\text{In}(\text{B}), \text{Go}(\text{C})) = \text{In}(\text{c})$

4. Goal Test

The current state is City J.

5. Path Cost

The sum of the distances (edge costs) along the chosen path from A to J.

- iii. Discuss **TWO (2)** differences between a **search tree** and a **state space**. (NATALIE ANG XI Q)

Search Tree

A tree structure generated by a search algorithm that shows the order in which states are explored.

The same state may appear multiple times because it can be reached through different paths during the search process.

State Space

The complete set of all possible states and actions of a problem, usually represented as a graph.

Each state appears only once because it represents every unique state and all possible connections between the states.

- iv. Construct the resulting **search tree** for **breadth-first search**. (RANAN GOH ZHI LIN)

A > B > C > E > J

- v. Discover both the **sequence of nodes in the frontier** from the beginning until a goal node is found, and **node expansion**. (RONAN CHAN A/L HARICHAN)

Expanded Node	Frontier	Reached(Frontier + Expanded)
	A	A
A	B,F	B,F
B	F,C,D	B,F,C,D
F	C,D,G,H	F,C,D,G,H
C	D,G,H,E	C,D,G,H,E
D	G,H,E	D,G,H,E
G	H,E,I	G,H,E,I
H	E,I	H,E,I
E	I, J(early goal test until here)	E,J

- vi. Identify the **solution**. (WONG JIA JIAN)

A,B,C,E,J

- vii. Calculate the **cost** of the solution. (Yap Jia Yu)

17

- viii. Evaluate the **performance of breadth-first search in solving this path-finding problem** in terms of completeness, optimality, time efficiency, and space efficiency where:

- Time efficiency is measured in terms of **the number of nodes expanded** and,
- Space efficiency is measured in terms of **the maximum size of frontier required** (**Remark:** the size is measured in byte, assuming that a node requires the storage of one byte). (ANG ZI CHEN)

Completeness- complete, BFS is complete when branching factor b is finite. The map has only 10 cities, so a solution is guaranteed to be found.

Optimality- not cost-optimal. BFS is only cost optimal when all the action costs are equal. The BFS return path cost is 17, however the cheapest path cost is 9.

Time efficiency, 8 nodes are expanded. The expansion order is A-B-F-C-D-G-H-E.

Space efficiency- 4byte. Since the biggest frontier holds 4 nodes at once and 1 node = 1 byte, the maximum memory needed is 4 bytes. For example frontier after expanding f which is [C,D,G,H].

Tutorial 5

Question 1

a) Given a map (see **Figure 1**) that contains 10 cities where the distance between any two cities is indicated by a number in brown, find the shortest path from city **A** to city **J** using a few strategies listed below:

- ☐ Depth-first search
 - Implemented with a **tree-like search algorithm**.
 - **Late goal test** is applied.
 - It is able to detect the presence of a looped path.
- ☐ Uniform-cost search
 - Implemented with a **graph search algorithm**.
 - **Late goal test** is applied.
 - If any 2 nodes have the same path cost, expansion will be done in alphabetical order.
- ☐ Iterative Deepening Search
 - Implemented with a **tree-like search algorithm**.
 - **Late goal test** is applied.

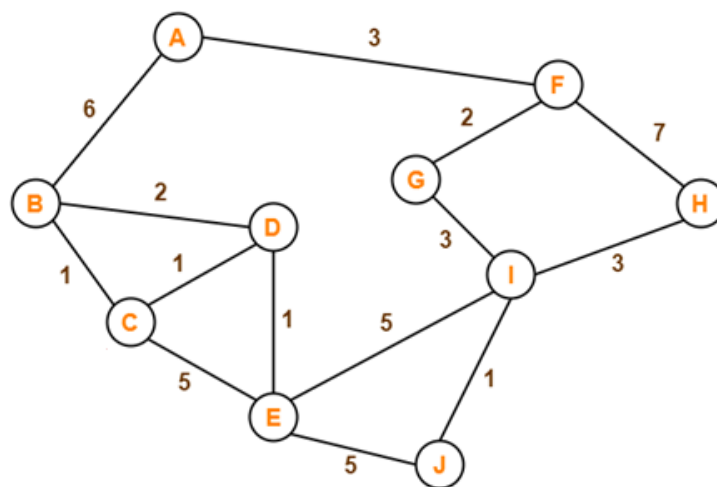


Figure 1. A map

Suppose nodes are expanded in alphabetical order, for each search strategy:

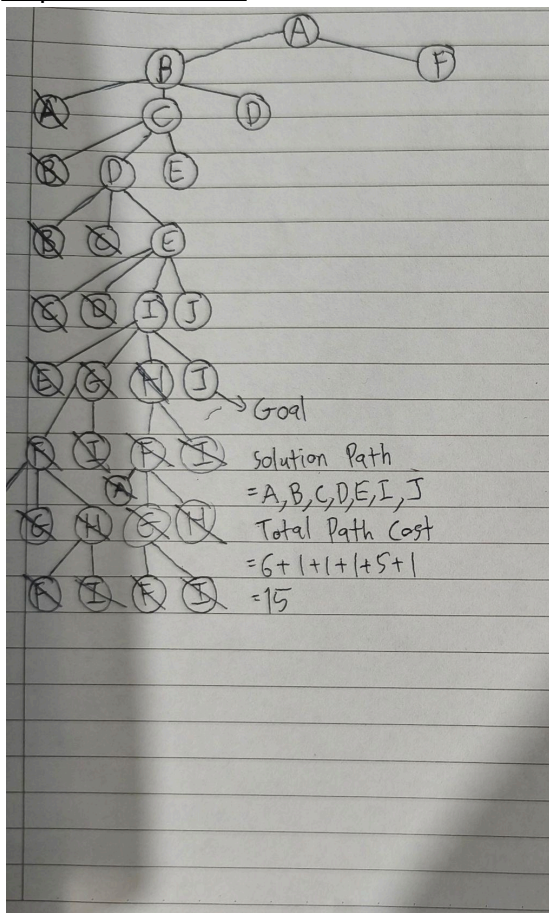
i. Construct the resulting **search tree**. (BEH CHUN YI)

A > F > G > I > J

ii. Discover both the **sequence of nodes in the frontier** and **node expansion**. (CHONG WEN KAI)

Answer wrong ady, here new one

Depth-First Search

	Expanded Node	Frontier
	-	A
	A	B, F
	B	A, C, D, F C, D, F
	C	B, D, E, D, F D, E, D, F
	D	B, C, E, E, D, F E, E, D, F
	E	C, D, I, I, E, D, F I, J, E, D, F
	I	E, G, H, J, J, E, D, F G, H, J, J, E, D, F
	G	F, I, H, J, J, E, D, F
	F	A, G, H, I, H, J, J, E, D, F H, I, H, J, J, E, D, F
	H	F, I, I, H, J, J, E, D, F I, H, J, J, E, D, F H, J, J, E, D, F
	H	F, I, J, J, E, D, F
	F	A, G, H, I, J, J, E, D, F G, H, I, J, J, E, D, F
	G	F, I, H, I, J, J, E, D, F J, J, E, D, F
	J	J, E, D, F

Uniform-Cost Search

NO: DATE:

Solution Path = A, F, G, I, J
 Total Path Cost = 0+3+2+3+1 = 9

Expanded Node	Frontier
-	A0
A0	B6, F3
F3	G5, H10, B6
G5	I8, H10, B6
B6	C7, D8, I8, H10
C7	E12, D8, I8, H10
D8	E9, I8, H10
I8	J9, E9, H10
E9	J9, H10
J9	H10

Iterative Deepening Search

iii. Identify the **solution**. (CHOW ZHEN WAI)

A, F, G, I, J

iv. Calculate the **cost** of the solution. (NEO JIA LER)

3+2+3+1=9

v. Evaluate its **performance** for solving this path-finding problem in terms of completeness, optimality, time efficiency and space efficiency, where:

- ❑ Time efficiency is measured in terms of **the number of nodes expanded** and,
- ❑ Space efficiency is measured in terms of **the minimum size of frontier required** (**Remark:** the size is measured in byte, assuming that a node requires the storage of one byte). (CLEMENT HO JUN HAO)

Completeness-Yes, it is completed, since the graph is finite which have only 10 nodes and 13 edges

Cost optimal-Yes it is cost optimal because it needs 9 path cost, another path like A>F>H>I>J needs 14 path cost

Time efficiency- 13 nodes expanded. Following the rules of Uniform-Cost Search (UCS) with a late goal test and tie-breaking by alphabetical order, the nodes are expanded in the following order: A, B, C, D, E, I, G, F, H, H, F, G, J

Space efficiency-It is required 1 byte of storage, The peak/maximum size of the frontier during this search is 1 node, meaning the required storage is 1 byte,

- vi. Among the four search strategies (including breadth-first search), identify **ONE (1)** best strategy for execution. Justify your answer. (ENG ZI SHAN)

Uniform-Cost Search (UCS)

- The graph has different path costs, not equal costs.
- UCS always expands the node with the lowest accumulated path cost ($g(n)$).
- Therefore, UCS is guaranteed to find the shortest (lowest-cost) path from A to J.
- BFS only guarantees the shortest path when all edge costs are equal, which is not true here.
- DFS is not optimal because it may go deep into an expensive path before finding the goal.
- Iterative Deepening Search (IDS) is also not cost-aware, it only considers depth, not distance

vii. Discuss **TWO (2)** differences between the following pairs in terms of their mechanism:

a. Breadth-first search vs. depth-first search (LEE NI)

1. Search mechanism

- Breadth-first search expands the root node first. Then, all the successors of the root node are expanded next. Their successors are expanded next, and all nodes at a given depth are expanded before any nodes at the next level.
- Depth-first search always expands the deepest node in the current frontier of the search tree. The search proceeds immediately to the deepest level of the search tree before exploring other branches.

2. Implementation

- Breadth-first search is implemented using a FIFO (First-In, First-Out) queue for the frontier.
- Depth-first search uses a LIFO (Last-In, First-Out) queue (stack), in which the most recently generated node is chosen for expansion.

b. Breadth-first search vs. uniform-cost search (FIONNA CHAN PUI YEE)

- **Evaluation Metric:** **BFS** expands nodes level by level based on **tree depth** (edge count), whereas **UCS** expands nodes based on **path cost**($g(n)$).
- **Cost Handling:** **BFS** assumes all step costs are equal (unweighted edges) to guarantee optimality. **UCS** handles varying, non-negative step costs (weighted edges) by using a priority queue to always expand the node with the lowest total path cost first.

Tutorial 6

Question 1

a) With the use of a map (see **Figure 1**) that contains 10 cities (where the distance between any two cities is indicated by a number in brown and heuristic values are numbers in red) as a state space, find the shortest path from city **A** to city **J** using a few strategies listed below:

- ☐ Greedy Best-First Search
 - Implemented with **tree-like search algorithm**.
 - **Late goal test** is applied.
 - **Assumption**: It is able to detect the presence of a looped path.
- ☐ A* Search
 - Implemented with a **graph search algorithm**.
 - **Late goal test** is applied.

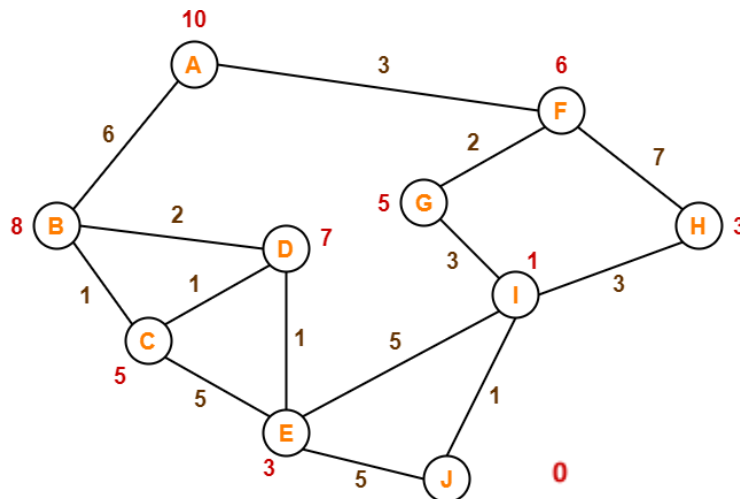


Figure 1. A map

i. Differentiate **informed search** from **uninformed search** in **THREE (3)** ways.

(JONG HUI XUAN)

Informed Search: Uses a heuristic function to estimate the cost or distance from the current state to the goal. Highly efficient with a smaller search space, leading to faster solutions and lower memory usage. Example A* Search, Greedy Best-First Search

Uninformed Search: Has no information about the goal distance and only knows how to generate successors and identify a goal state. Less efficient, time-consuming, and can suffer from high memory consumption because it explores many irrelevant paths. Example Breadth-First Search (BFS), Depth-First Search (DFS).

ii. Suppose nodes are expanded in alphabetical order, for each search strategy:

a. Construct the resulting **search tree**. (PANG YI WEN)

--

b. Identify both the **sequence of nodes in the frontier** and **node expansion**. (TAN HAOKIT)

Greedy Best-First Search

Expanded node	Frontier
	A10
A10	F6, B8
F6	H3, G5, B8
H3	I1, G5, B8
I1	J0, E3, G5, G5, B8
J0	E3, G5, G5, B8

A* Search

Expanded node	Frontier
	A10
A10	F9, B14
F9	G10, H13, B14
G10	I9, H13, B14

I9	J9, H13B14
J9	H13, B14, E16

- c. Identify the **solution**. (KHOO CHEE YOUNG)

A,F,G,I,J

- d. Calculate the **cost** of the solution. (NG YIT KAI)

Greedy Best-First Search ($A > F > H > I > J$)

Cost = $3 + 7 + 3 + 1 = 14$

A* Search ($A > F > G > I > J$)

Cost = $3 + 2 + 3 + 1 = 9$

- e. Evaluate its **performance** for solving this path-finding problem in terms of completeness, optimality, time and space efficiency.
- Time efficiency is measured in terms of **the number of nodes expanded until a goal node is discovered** and,
 - Space efficiency is measured in terms of **the maximum size of frontier required** (Remark: the size is measured in bytes, assuming that a node requires the storage of one byte). (LEN CHUN HOE)

Time efficiency

Greedy Best-First Search ($A > F > H > I > J$) = 5 nodes

A* Search ($A > F > G > I > J$) = 5 nodes

Space efficiency

Greedy Best-First Search = 3 nodes

A* Search = 3 nodes

- iii. Between the **TWO (2)** informed search strategies, select **ONE (1)** for execution. Justify your answer. (LEONG ZHENG YANG)

Greedy Best First Search

A* Search

select A* Search

-evaluates nodes using $f(n) = g(n) + h(n)$, combining actual path cost with the heuristic estimate. This makes it both complete and optimal.

- iv. Between the **SIX (6)** search strategies, select **ONE (1)** for execution. Justify your answer. (LUI FANG YEE)

A* search is selected for execution.

The reason is that even though both uniform-cost search and A" search are complete and cost-optimal, A* search has higher time efficiency, and hence it is able to return a solution faster

- v. Discuss **TWO (2)** differences between the following pairs in terms of their mechanism:

- a. Greedy best-first search vs. A* search (NATALIE ANG XI QI)

Greedy best-first search

A>F>H>I>J, with cost 14

A* search

A>F>G>I>J, with total cost 9

- b. Greedy best-first search vs. uniform-cost search (RANAN GOH ZHI LIN)

Evaluation $f(n)$: (GBFS) $h(n)$ – estimated cost to goal
(UCS) $g(n)$ – actual cost from start to current node
Optimality: (GBFS) Not guaranteed
(UCS) Guaranteed optimal
Completeness: (GBFS) Not complete
(UCS) Complete

b) A* search relies on the use of **heuristics** in solving the path-finding problem. However, whether it is cost-optimal or not, it depends on **TWO (2) properties** of the heuristic utilized.

i. Name the **TWO (2)** properties and briefly describe each of them. (RONAN CHAN A/L HARICHAN)

1) Admissibility

- a heuristic is admissible if it never estimates the true cost to reach the goal from any given node.
- controls $h(n)$

2) Consistency

- A heuristic is consistent if for every edge from node n to a successor, the estimated cost to the goal is less than or equal to the step cost plus the estimated cost to the goal for the successor.
- controls how $h(n)$ changes relative to $g(n)$ and step cost, ensuring smooth graph-search behaviour

ii. Prove the following **TWO (2)** statements are **TRUE**:

o A* search is **cost-optimal** if it has an **admissible heuristic**. (WONG JIA JIAN)

- Let the optimal solution cost be C^*
- For any node n on an optimal path: $g(n) + h(n) \leq g(n) + h^*(n) = C^*$
- Therefore, $f(n) \leq C^*$
- A* always has at least one node on an optimal path whose $f(n)$ value is no greater than C^*

- Suppose A^* select a goal node with cost $C > C^*$. There must still be an unexpanded node n on the optimal path with: $f(n) \leq C^* < C$
- A^* would select n instead of the more expensive goal because it chooses the node with the smallest $f(n)$
- This is contradiction
- Therefore, A^* cannot select a suboptimal goal before an optimal one
- Hence, A search is cost-optimal when the heuristic is admissible

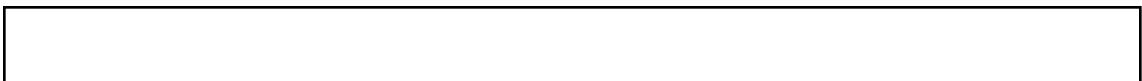
- A^* search is **cost-optimal** if it has a **consistent heuristic**. (Yap Jia Yu)

- Consistency $\Rightarrow f$ is non-decreasing along any path: $f(n') = g(n) + c(n, a, n') + h(n') \geq g(n) + h(n) = f(n)$.
- Since A^* expands nodes in non-decreasing f order, each node is first reached via its cheapest path (graph search never needs to reopen nodes).
- When the goal is expanded, $f(\text{goal}) = g(\text{goal})$ ($h=0$), and it's $\leq$ every other frontier node's f — so no cheaper path to the goal exists.
- Therefore A^* graph search is optimal given consistent h .
- Consistency is the safer condition for graph search since plain admissibility alone can let graph search lock in a suboptimal g -value for a node.

Tutorial 7

Question 1

- a) A **state-space landscape** is a graphical representation of different states of a problem reachable by a search algorithm.
- i. Draw a state-space landscape in which its elevation corresponds to an objective function. (ANG ZI CHEN)



- ii. A hill-climbing search can be illustrated with the use of a state-space landscape. Identify and name **THREE (3)** points/areas in which a hill climbing search can get stuck. Then, briefly describe each of them. (BEH CHUN YI)

- iii. Describe **TWO (2)** solutions to the problem mentioned in **Question 1 a) ii)**. (CHONG WEN KAI)

- b) **Course scheduling** is a common problem in many universities, which is concerned with developing a timetable that illustrates a number of courses assigned to the classrooms by taking the constraints of university stakeholders such as students, lecturers, and departments and classroom capacities into consideration.

- i. The **University Course Scheduling Problem** (UCSP) is an **optimization problem**. Identify whether this statement is true or false. Justify your answer. (CHOW ZHEN WAI)

- ii. Which type of strategies is more practical in solving UCSP, **classical search strategies**, or **local search strategies**? Justify your answer. (NEO JIA LER)

- iii. Hill climbing is one of the popular local search strategies. Briefly discuss how its variants listed below will work towards reaching the goal state. (CLEMENT HO JUN HAO)

- ☐ **Stochastic hill climbing**
- ☐ **First-choice hill climbing**
- ☐ **Random-restart hill climbing**

- iv. Suppose you are planning to solve the above-mentioned UCSP using a hill climbing search, determine whether the basic steepest ascent version or its variants should be used. Justify your answer. (ENG ZI SHAN)

Question 2

- a) Suppose a one-column one-queen formulation is used in a **4-Queen problem**, and the heuristic cost function h is defined as the **number of pairs of queens that are attacking each other, either directly or indirectly**. Answer the following questions:
- i. Calculate the total number of possible states for the 4-Queen problem. (LEE NI)

- ii. Calculate the number of successors for each of the 4-Queen states. (FIONNA)

- iii. Calculate the h -value of the initial state of the problem provided in **Figure 1**. (JONG HUI XUAN)

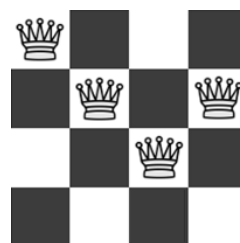


Figure 1. The initial state

- iv. Fill up the other squares with the proper h -values. (PANG YI WEN)

- v. Conduct a **simulation** on how the problem is solved using the **basic steepest ascent version of hill climbing search**. (TAN HAO KIT)

- vi. Describe various states the hill-climbing search would encounter using a **hill-climbing landscape**. (KHOO CHEE YOUNG)

- b) Suppose a heuristic cost function h is applied to solve an **8-puzzle problem**, where h is the **sum of the distances of the 8 tiles from their goal positions** (i.e., the sum of the horizontal and vertical distances only), given an initial state and a goal state provided in **Figure 2**, answer the following questions:

	1	3
4	2	5
7	8	6

Initial State

1	2	3
4	5	6
7	8	

Goal State

Figure 2. Initial state and goal state of a 8-puzzle problem

- i. Calculate the h -values of the initial state and the goal state. (**Remark:** Show the steps). (NG YK)

- ii. Calculate the h -values of the resulting states as a result of applying all available actions in the initial state. (**Remark:** Show the steps). (LEN CH)

- iii. Conduct a **simulation** on how the problem is solved using the **basic steepest ascent version of hill climbing search**. (**Remark:** The h -values of all states must be displayed.) (LEONG ZY)

- iv. Conduct a **simulation** on how the problem is solved using **local beam search** with:

- ☐ Beam width, $k = 1$. (LUI FY)

- ☐ Beam width, $k = 2$. (**Remark:** If more than one neighbor has the same h -value, any neighbor with the lowest tile number will be selected, but any neighbor results from any immediate reverse move will not be considered.) (NAT)

- v. Discuss **TWO (2)** differences between **random-restart hill climbing** and **local beam search**. (RANAN)

Question 3

- a) Suppose an Evolutionary Algorithm (EA) uses a series of 8-digit string to represent the state of a particular problem in the form of $X_i = \mathbf{ABCDEFGH}$ where i is a running number and each digit is in the range of 0 and 9 inclusive, its fitness function is $f(\mathbf{X}) = (\mathbf{A} + \mathbf{B}) - (\mathbf{C} + \mathbf{D}) + (\mathbf{E} + \mathbf{F}) - (\mathbf{G} + \mathbf{H})$, and the initial randomly generated states are represented by the following 4 strings, answer the following questions:

$$X_1 = 74235341$$

$$X_2 = 32567741$$

$$X_3 = 85390164$$

$$X_4 = 26730459$$

- i. Calculate the fitness of each string (show the steps). Then, arrange them in a descending order based on their fitness scores. (RONAN)

- ii. Perform the following crossover operations:

1. Cross the fittest two strings using a single-point crossover (at the middle point).

(WONG JJ)

2. Cross the second and third fittest strings using a two-point crossover (point's B and F). (YAP JY)

3. Cross the first and third fittest strings using a uniform crossover. (ANG ZC)

- iii. Suppose the crossover operations mentioned in Question 1 a) ii) are completed, and the new population consists of the six offspring strings now. Calculate the fitness of the new population again (show the steps). Then, compare them with their 4 parent strings, and identify whether the overall fitness of the offspring strings has improved. Justify your answer. (BEH CY)

- iv. Repeat Step ii) and iii) of Question 1 for 3 to 5 times. Compare the overall fitness of the offspring strings obtained, and identify whether the overall fitness of the offspring strings has improved. Justify your answer. (CHONG WK)

- v. Identify the string representation of the optimal solution. Then, calculate its maximum fitness. (CHOW ZW)

- vi. Discuss the gap with between the overall fitness and the maximum fitness. (NEO JL)

- vii. Will the EA be able to find the optimal solution without having the mutation operation? Justify your answer. (CLEMENT)

Tutorial 8

Question 1

In Artificial Intelligence (A.I.), games are modeled as a search problem, also known as adversarial search problems.

- i. List some characteristics of commonly studied games in A.I. and briefly describe each of them. (ENG ZS)

- ii. Tic-Tac-Toe is a popular example of such games. List down the types of environment for this game. Justify your answer. (LEENI)

- iii. Games can be solved in four steps like how we solve the ordinary search problem. In the second step, a game is formulated with 6 important elements. Name and briefly describe the **SIX (6)** elements. Then, define Tic-Tac-Toe in terms of the six elements.

(FIONNA)

- iv. Tic-Tac-Toe is a two-player game. Explain why the two players are referred to as MAX and MIN. (JONG HX)

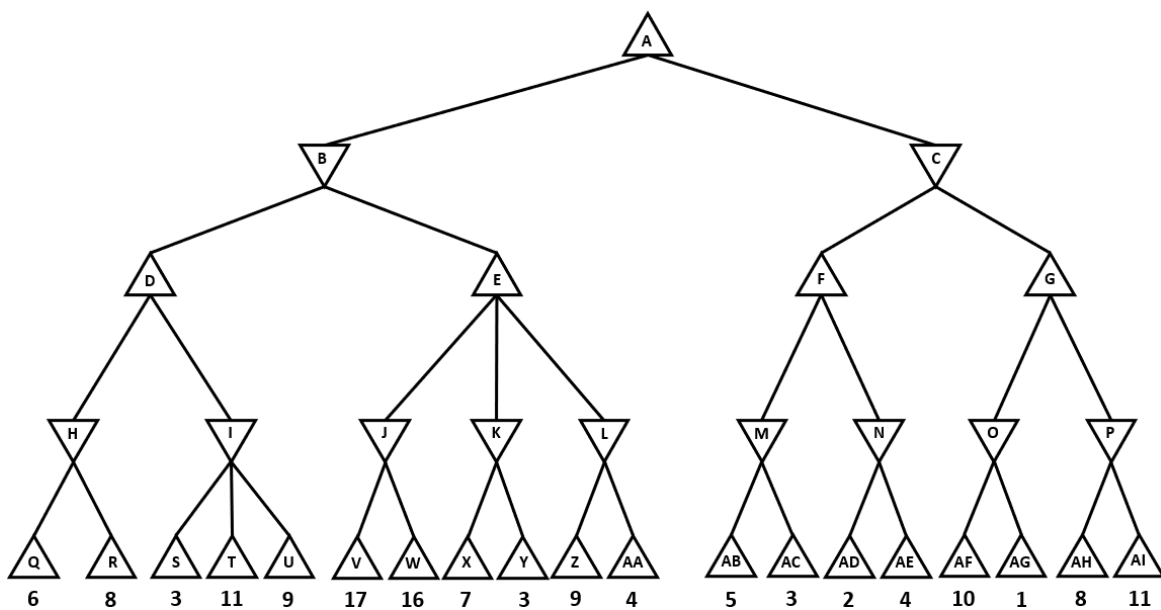
a) In a two-player game, suppose both MAX and MIN play optimally from the beginning until the end of the game, the optimal strategy can be determined by working out the minimax value of each state in a game tree.

i. Describe the minimax value. (PNAG YW)

ii. Describe the game tree. (TAN HK)

iii. Explain what will happen to MAX if MIN doesn't play optimally. (KHOO CY)

b) Answer the following questions based on the game tree given below. (NG YK)



i. Label each level to show which one represents MAX's and MIN's moves.

--

- ii. Identify the minimax value of each node from A to P.

--

- iii. Identify the best move for MAX.

--

Tutorial 9

Question 1

a) According to Wikipedia, alpha-beta pruning is a search algorithm that seeks to decrease the number of nodes that are evaluated by the minimax algorithm in its search tree.

- i. Alpha-beta pruning gets its name from two parameters: α or β . Briefly describe each of them. (LEN CH)

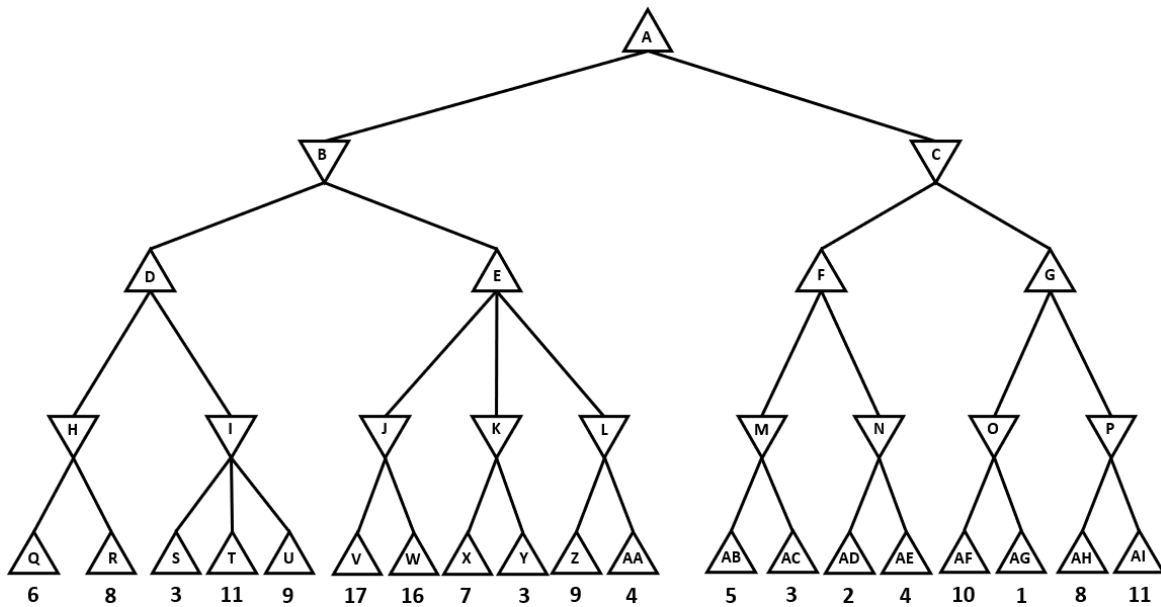
- ii. Explain how the minimax algorithm can be optimized by alpha-beta pruning. (LEONG ZY)

- iii. When alpha-beta pruning is applied to the minimax algorithm, explain what will be affected and what will not be affected. (LUI FY)

- iv. Discuss ONE (1) advantage and disadvantage of using alpha-beta pruning. (NAT)

- v. Compare alpha-beta pruning and forward pruning. (RANAN)

b) Answer the following questions based on the game tree given below.



- i. Apply the minimax algorithm to the game tree with alpha-beta pruning in left-to-right order. Show how the expansion of the game tree and pruning are done step-by-step.

(RONAN)

- ii. Then, identify nodes or branches that would be pruned by alpha-beta pruning. (Same student who did Q1b(i)) RONAN

- iii. Apply the minimax algorithm to the game tree with alpha-beta pruning again, but in right-to-left order. Show how the expansion of the game tree and pruning are done step-by-step. (WONG JJ)

- iv. Then, identify nodes or branches that would be pruned by alpha-beta pruning. (Same student who did Q1b(iii)) WONG JJ

--

- v. Compare the two minimax values computed at the root node and the nodes/branches pruned during the search operation. Discuss your findings. (YAP JY)

--

Question 2

- a) Briefly explain each of the **SIX (6)** techniques presented in **Table 1** as follows: (ANG ZC)

Table 1: Search-related techniques

Label	Name
A	minimax
B	Alpha-beta pruning
C	Cut-off search
D	Heuristic function
E	Transposition table
F	Lookup table

- b) Briefly explain how different combinations of the six techniques presented in **Table 1** will affect the search performance:

- i. A + B (BEH CY)

- ii. A + B + C + D (CHONG WK)

- iii. A + B + C + D + E (CHOW ZW)

- iv. A + B + C + D + E + F (NEO JL)

c) Cut-off search was proposed by Claude Shannon in 1950 in his research paper titled "**Programming a Computer for Playing Chess**" to tackle the practicality issues of the adversarial search problem.

i. Cut-off search can be implemented by controlling the amount of search in the following **TWO (2)** ways:

- ☐ Setting a fixed limit
- ☐ Iterative deepening search.

Briefly describe each of them. Then, identify a better approach supported by justification. (CLEMENT)

ii. The following **THREE (3)** issues arise in the implementation of cut-off search:

- ☐ Approximation errors
- ☐ Non-quiesscence positions
- ☐ Horizon effects

Briefly describe and provide ONE (1) example for each of the issues. Then, suggest a solution to each of the issues. (ENG ZS)

Tutorial 10

Question 1

a) Suppose $X_i = \{ X_1, X_2, X_3 \}$ represents three students from the DCS programme who are qualified for the Best Student Award. The domain of each variable is the set $D_i = \{0, 1\}$. For instance, when $X_1 = 1$, it means that X_1 will be awarded the Best Student Award; when $X_1 = 0$, it means that X_1 will not be awarded. Write a constraint to represent each of the following in two ways: 1) an abstract relation and 2) an enumeration of the members of the relation:

- i. At least two of the three students must be awarded (LEENI)

--

- ii. At most one of the three students can be awarded (FIONNA)

--

- iii. Only one in between student 1 and student 3 is awarded (JONG HX)

--

- iv. Student 2 and Student 3 must be awarded or not awarded at the same time (PANG YW)

--

- v. The three students cannot all be awarded, there must be at least one that is not awarded (TAN HK)

--

- vi. If student 2 is awarded, then student 1 must be awarded; but if student 2 is not awarded, then there is no restriction on student 1. (NG YK)

Question 2

Table 1: Availability of courses

No.	Course Code	Course Name	Time
1	AACS2123	Computer Game Development	8:00am – 10:00am
2	AACS2383	Introduction to Data Mining	9:00am – 11:00am
3	AACS3013	Database Development and Applications	10:30am – 12:30pm
4	AACS3064	Computer Systems Architecture	11:00am – 1:00pm
5	AACS3273	Fundamentals of Artificial Intelligence	11:00am – 1:00pm

Table 2: Availability of lecturers

No.	Lecturer	Courses
1	Alex	1, 3, 4
2	Billy	2, 3, 4, 5
3	Carolyn	1, 2, 4

- a) The new semester is about to begin, and you are in charge of assigning 5 computer science courses to 3 academic staff based on the information provided in **Table 1** and **Table 2**. Suppose courses that can be taught by the academic staff are listed above and each of them can only be teaching one course at a time: (LEN CH)
- i. Formulate this problem as a CSP problem in which there is one variable per class, stating the domains (after having already enforced any unary constraints), and constraints.

- ii. Draw the constraint graph associated with your CSP.

- iii. Show the domains of the variables after running arc-consistency on this initial graph (after having already enforced any unary constraints).

- iv. Give one solution to this CSP.

- b) Redo **Question 2a i) to iv)** based on the information provided in **Table 1** and **Table 3**.
(LEONG ZY)

Table 3: Availability of lecturers

No.	Lecturer	Courses
1	Alex	3, 5
2	Billy	2, 3, 4, 5
3	Carolyn	1, 2, 4

- i. Formulate this problem as a CSP problem in which there is one variable per class, stating the domains (after having already enforced any unary constraints), and constraints.

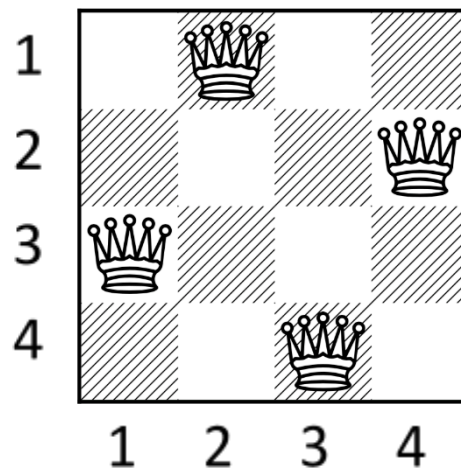
- ii. Draw the constraint graph associated with your CSP.

- iii. Show the domains of the variables after running arc-consistency on this initial graph (after having already enforced any unary constraints).

- iv. Give one solution to this CSP.

Question 3

- a) Suppose the formulation of one queen in each column is applied, model a 4-Queens problem as a CSP, and write down the variables, domains, and constraints. (LUI FY)



Tutorial 11

Question 1

- a) The new semester is about to begin, and you are in charge of assigning 3 computer science courses to 2 academic staff based on the information provided in **Table 1** and **Table 2**.

Table 1: Availability of courses

No.	Course Code	Course Name	Time
1	AACS2123	Computer Game Development	8:00am – 10:00am
2	AACS2383	Introduction to Data Mining	9:00am – 11:00am
3	AACS3013	Database Development and Applications	9:30am – 11:30am

Table 2: Availability of lecturers

No.	Lecturer	Courses
1	Alex	1, 2, 3
2	Billy	1, 2, 3

- i. Formulate this problem as a CSP problem in which there is one variable per class, stating the domains (after having already enforced any unary constraints), and constraints. (NATALIE)

- ii. Draw the constraint graph associated with your CSP. (Same student who did Q1a(i))NAT

- iii. Show the domains of the variables after running arc-consistency on this initial graph (after having already enforced any unary constraints). (Same student who did Q1a(i))NAT

- iv. **"The constraint graph is arc-consistent after the implementation of AC-3."**

State your agreement with the above statement. Justify your answer with the use of an example. (Same student who did Q1a(i))NAT

- v. Define a solution to CSP. (Same student who did Q1a(i))NAT

- vi. **"There is no solution to this CSP."**

State your agreement with the above statement. Justify your answer. (RANAN)

- vii. Make a conclusion based on the two statements provided in **Question 1a iv)** and **Question 1a vi)**. (Same student who did Q1a(vi))RANAN

- viii. ***"AllDiff constraint is a possible solution to the issue stated in Question 1a vii)".***

State your agreement with the above statement. Justify your answer (**Remark:** if your answer is yes, explain how the solution works). (Same student who did Q1a(vi))RANAN

- ix. Briefly explain how can **Alldiff** constraint can easily detect the presence of any inconsistency. (Same student who did Q1a(vi))RANAN

- x. Reformulate the problem given in **Question 2a)** as a CSP problem with the use of the **Alldiff** constraint. (Same student who did Q1a(vi))RANAN

Question 2

- a) Sudoku is a popular puzzle game based on the logical placement of numbers. It has introduced millions of people to **Constraint Satisfaction Problem (CSP)**, although they may not recognize it. A classical Sudoku has a 9×9 grid consists of 81 squares, some of which are initially filled with digits from 1 to 9. The objective is to fill the 9×9 grid with digits so that each column, each row, and each of the nine 3×3 sub-grids that compose the grid contain all of the digits from 1 to 9.

- i. Formulate the 9×9 Sudoku as a CSP and identify the variables, domains for each variable and the constraints using the **Alldiff** global constraint. (RONAN)

- ii. A sample Sudoku puzzle is provided in **Figure 1**.

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7								8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

Figure 1: Sudoku

Suppose **Alldiff** constraints have been expanded into binary constraints so that AC-3 algorithm (i.e., the most popular algorithm for enforcing arc consistency) can be applied directly, explain how the following variables are made arc-consistent step-by-step.

(WONG JJ)

- ☐ I4
- ☐ A4
- ☐ E4

Tutorial 12

Question 1

a) Similar to classical search problems and optimization problems, CSPs can also be solved with the use of search algorithms. These search algorithms can be divided into backtracking search and local search.

i. Briefly describe the difference between backtracking search and local search. (YAP JY)

ii. Backtracking search is a variant of depth-first search. Differentiate it from depth-first search. (ANG ZC)

iii. Unlike informed search, backtracking search doesn't rely on a domain-specific heuristic. Instead, it makes use of domain-independent heuristics. Differentiate domain-specific heuristics from domain-independent heuristics. (BEH CY)

iv. Identify **FOUR (4)** different ways that can affect the performance of backtracking search. (CHONG WK)

v. Explain how backtracking search can be improved with the use of the following combinations:

a) Backtracking + Forward Checking (NEO JL)

b) Backtracking + Forward Checking + Maintaining Arc Consistency (MAC) Algorithm
(CLEMENT)

- vi. **Figure 1** shows a constraint graph. Highlight the scope of backtracking, forward checking and forward checking + MAC algorithm: (ENG ZS)
- ☐ Arcs represent constraints
 - ☐ Nodes represent variables
 - Nodes in grey represent variables assigned a value
 - Nodes in black represent the current variable that has just been assigned a value
 - Nodes in white represent variables not assigned any value yet

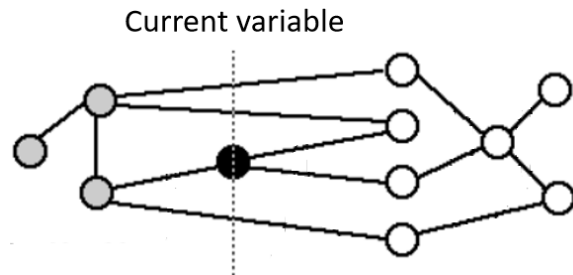


Figure 1: Constraint graph

- b) The new semester is about to begin, and you are in charge of assigning 5 computer science courses to 3 academic staff based on the information provided in **Tables 1** and **2**.

Table 1: Availability of courses

No.	Course Code	Course Name	Time
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1	AACS2123	Computer Game Development	8:00am – 10:00am
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Table 2: Availability of lecturers

No.	Lecturer	Courses
1	Alex	1, 3, 4
2	Billy	2, 3, 4, 5
3	Carolyn	1, 2, 4

- i. Suppose this problem is formulated as a CSP, and by referring to your answer for **Tutorial 10 Question 2a i)**, simulate how the backtracking algorithm works in searching a solution for this CSP. To answer this question, you may refer to a handout named "**Simulation of Backtracking and AC-3**". (LEENI)

- ii. Simulate backtracking with the use of the following heuristic separately. Then, discuss how the usage of the heuristic affects the performance backtracking search:
- a) Minimum-Remaining-Values (MRV) heuristic (FIONNA)

- b) Degree heuristic (JONG HX)

- c) Least-Constraining-Value heuristic (PANG YW)

Question 2

- a) Given a map of Australia that shows each of its states and territories as illustrated in **Figure 2**, you are asked to color all the states using either **Red**, **Green**, or **Blue** in such a way that no neighboring regions have the same color. Suppose backtracking is applied with a fixed variable ordering **WA, V, NT, Q, NSW, T, SA** and **{ WA = Red, V = Blue, NT = Blue, Q = Red, NSW = Green, T = Red }** is given as the partial assignment and the 3 colors are assigned in the sequence of **Red**, **Green**, or **Blue**, answer the following questions:



Figure 2: A map of Australia

- i. The next assignment is **SA**. Since there is no legal value left to be assigned to it, the algorithm will backtrack. Identify a decision point where the algorithm should backtrack to. Justify your answer. (TAN HK)

- ii. "The chronological backtracking approach stated in Question 2a i) is not so effective."

State your agreement with a statement. Justify your answer. (KHOO CY)

- iii. **Backjumping** is an alternative approach to the backtracking mechanism stated in **Question 2a i)**. Discuss how backjumping is different from the **chronological backtracking approach**. (NG YK)

- iv. Backjumping can be implemented with the use of a **conflict set**. Identify all elements of **conflict set** for the following variables:

a) **WA** (LEN CH)

b) **V** (LEONG ZY)

c) **NT** (LUI FY)

d) **Q** (NAT)

e) **NSW** (RANAN)

f) **SA** (RONAN)

- v. In the case of no legal values to be assigned to **SA**, briefly explain how **backjumping** works with the use of the **conflict sets** provided in **Question 2a iv)**. (WONG JJ)

- vi. Discuss the effectiveness of backjumping in comparison with simple backtracking in this case. (YAP JY)

Tutorial 13

Question 1

a) Identify whether the following statements are **True**, **False** or **not a valid statement**. Justify your answer if the it is not true.

i. An integer is a rational number (ANG ZC)

ii. Let x be a real number (BEH CY)

iii. $5 + 5 = 10$ and $7 < \sqrt{50}$ (CHONG WK)

iv. 11 is an even integer and $64^{-1/6} = \frac{1}{2}$ (CHOW ZW)

v. 11 is an even integer or $64^{-1/6} = \frac{1}{2}$ (NEO JL)

vi. $(5 + 5 \neq 11) \Rightarrow 7 < \sqrt{50}$ (CLEMENT)

vii. $(5 + 5 \neq 11) \Rightarrow 7 > \sqrt{50}$ (ENG ZS)

- b) Let **P** stand for the proposition "**I took part in the competition**" and **Q** stands for "**I won 2nd place.**". Express the following as natural English sentences:

i. $\neg P$ (LEENI)

ii. $P \wedge Q$ (FIONNA)

iii. $P \Rightarrow Q$ (JONG HX)

iv. $\neg P \Rightarrow \neg Q$ (PANG YW)

v. $\neg P \vee (P \wedge Q)$ (TAN HK)

- c) Formalise the following in terms of atomic propositions **H**, **C**, and **S**. But you have to first make it clear how they correspond to the English text.

- i. The hall is crowded with people, but the celebrity has not arrived. (KHOO CY)

--

- ii. Celebrity has not arrived, and finding a seat in the hall is easy, but the hall is crowded with people. (NG YK)

--

- iii. If the hall is crowded with people, then finding a seat is easy if and only if the celebrity has not arrived. (LEN CH)

--

- iv. It is not easy to find a seat in the hall, but the celebrity has not arrived and the hall is crowded with people. (LEONG ZY)

--

- v. Finding a seat in the hall is not easy whenever a celebrity has arrived and the hall is crowded with people. (LUI FY)

--

Question 2

- a) Fill in the missing columns in the truth table above. (NAT)

P	Q	$P \Rightarrow Q$	$P \Leftrightarrow Q$	$\neg Q \Rightarrow \neg P$	$\neg P \vee Q$	$(\neg P \wedge \neg Q) \vee (Q \wedge P)$

T	T	T	T			
T	F	F	F			
F	T	T	F			
F	F	T	T			

b) In the truth table above, which statements are logically equivalent? (RANAN)

c) Construct a truth table for each of the following propositions and identify if it is a tautology:

i. $P \wedge Q \Rightarrow P$ (RONAN)

ii. $P \wedge (P \Rightarrow Q) \Rightarrow Q$ (WONG JJ)

iii. $(P \vee Q) \wedge \neg P \wedge \neg Q$ (YAP JY)

iv. $P \vee \neg(P \wedge Q)$ (ANG ZC)

v. $(P \Rightarrow Q) \Leftrightarrow (\neg P \Rightarrow \neg Q)$ (BEH CY)

Question 3

a) Convert the following sentences to conjunctive normal form.

i. $P \Leftrightarrow Q$ (CHONG WK)

ii. $(P \Rightarrow Q) \Rightarrow R$ (CHOW ZW)

iii. $P \Rightarrow (Q \Rightarrow R)$ (NEO JL)

iv. $(P \Rightarrow (Q \Rightarrow R)) \Rightarrow (P \Rightarrow (R \Rightarrow Q))$ (CLEMENT)

Special Words



**GOOD LUCK
ON EXAMS!**

**"Do the best you can
until you know better.
Then when you know
better, do better."
-Maya Angelou**



